

Experiment 3-Medium Wind: Summary Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1988	2.268	1.316	4.508	36.0%	71.0%	84.1%	13.812	28.068	8.630	21.079	0.558	1.282	1.420
Proposed	1984	2.122	0.975	3.805	51.3%	81.2%	88.4%	15.190	28.788	7.504	16.421	0.532	1.264	1.466
PPO-GRU	1984	2.680	1.185	8.280	42.5%	74.6%	84.7%	14.007	28.654	6.997	15.409	0.453	1.174	1.415
MPC	1035	5.681	3.061	15.400	4.1%	18.1%	48.7%	2.473	3.694	1.384	2.557	0.141	0.144	0.166

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1988	99.4%	12	0.6%	0.6%	12	100.0%	0	0
Proposed	2000	1984	99.2%	16	0.8%	0.8%	16	100.0%	0	0
PPO-GRU	2000	1984	99.2%	16	0.8%	0.8%	16	100.0%	0	0
MPC	2000	1035	51.8%	965	48.2%	48.2%	965	100.0%	0	0

Cross-Case Check: PPO/Proposed/PPO-GRU on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	965	956	99.1%	194	20.3%	2.497
Proposed	965	956	99.1%	129	13.5%	2.306
PPO-GRU	965	955	99.0%	152	15.9%	2.889